

Hoonki Yeo

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RESEARCH INTERESTS

Stochastic, derivative-free, and large-scale optimization, with applications to machine learning and data-driven decision problems.

EDUCATION

Georgia Institute of Technology

08/2025 – 12/2026 (Expected)

Master of Science in Operations Research

Atlanta, GA

- Relevant Coursework: Convex Optimization, Linear Optimization, Simulation, Theoretical Statistics, High-Dimensional Data Analytics, Computational Data Analytics

University of Wisconsin–Madison

08/2019 – 08/2022

Bachelor of Science in Data Science

Madison, WI

- GPA: 3.84/4.00
- Honors and Awards: Graduated with Distinction, 4× Dean's List

TEACHING & LEADERSHIP EXPERIENCE

Peer Tutor

01/2020 – 05/2021

University of Wisconsin–Madison

Madison, WI

- Provided one-on-one tutoring support for STAT 301 and STAT 309 (statistics and probability).

Republic of Korea Army

10/2016 – 07/2018

Sergeant (TOD Operator)

Yeoncheon, South Korea

- Operated and maintained thermal/optical observation systems and led a small team responsible for equipment readiness and training new personnel.

RESEARCH EXPERIENCE

Graduate Researcher

03/2026 – Present

Georgia Institute of Technology, advised by Prof. Sen Na

Atlanta, GA

- Conducting research in stochastic optimization under Prof. Sen Na, with a focus on application-oriented extensions of derivative-free constrained optimization methods.
- Currently formulating equality-constrained black-box optimization problems arising in machine learning model tuning, where continuous tuning parameters affect training and validation performance.

PROFESSIONAL EXPERIENCE

Data Scientist

12/2023 – 02/2025

Aivelabs

Seoul, South Korea

- Built an AI-driven CRM platform for customer value forecasting, integrating data pipelines, model training, and production scoring to support retention targeting.
- Designed and deployed a personalized recommendation system to rank outreach candidates and improve targeting for marketing/sales operations.
- Defined key business KPIs and implemented BI dashboards to monitor sales/funnel metrics, collaborating with cross-functional teams to deliver actionable insights for client decisions.

AI Research Bootcamp Participant (Time Series)

07/2023 – 09/2023

LG AI Research

Seoul, South Korea

- Selected for and completed an intensive bootcamp on multivariate time-series analysis and forecasting.
- Led a five-member team in the capstone hackathon and placed **23rd out of 747 teams (Top 3%)**.

Data Analyst

10/2022 – 05/2023

Retentics

Seoul, South Korea

- Maintained data processing and validation pipelines for 10+ clients, ensuring consistent schemas and alignment with internal product standards.
- Delivered ad hoc analyses for 5+ clients to support operational decisions and strategy, emphasizing clear assumptions and reproducible workflows.
- Developed an internal Python package to standardize data preprocessing, reducing processing time by **30%**.

SELECTED PROJECTS

AACE: AI-Powered CRM Platform Development

12/2023 – 02/2025

AiveLabs

Seoul, South Korea

- Developed an LTV prediction model using probabilistic modeling (BG/NBD & Gamma–Gamma) to support customer value forecasting and retention strategies.
- Implemented a feedback loop using **multi-armed bandits (MAB)** to balance exploration–exploitation, resulting in a **6% improvement in response rate**.
- Built a customer segmentation pipeline using PCA and K-means clustering for targeted marketing and personalization.

Sales Forecasting for LG Household & Health Care

07/2023 – 09/2023

LG AI Research Hackathon

Seoul, South Korea

- Implemented a bidirectional LSTM forecasting model for 1,000+ products using a multivariate time-series dataset (20M+ records), outperforming ARIMA/SARIMA baselines.
- Tuned model performance via Bayesian hyperparameter optimization (**Optuna**) and ranked **23rd out of 747 teams (Top 3%)**.
- Explored predictive uncertainty estimation using stochastic dropout sampling at inference (MC Dropout).

Lake Mendota Ice-Cover Duration Forecasting

01/2022 – 02/2022

CS 540: Introduction to AI, Prof. Jerry Zhu

Madison, WI

- Automated the collection of 70 years of historical climate data from the Wisconsin State Climatology Office using **Selenium**.
- Estimated model parameters using **maximum likelihood estimation (MLE)** and trained a linear regression model for forecasting ice-cover duration.
- Evaluated predictive performance on a held-out period; results indicated the model captured long-term trends and aligned with 2022 season behavior.

Fetal Health Classification

09/2021 – 12/2021

STAT 451: Intro to ML, Prof. Sebastian Raschka

Madison, WI

- Trained multi-class classifiers (KNN, Random Forest, Histogram Gradient Boosting) and achieved **96% accuracy**.
- Conducted EDA and feature-importance analysis to identify key physiological indicators associated with fetal health status.
- Evaluated performance with cross-validation and confusion-matrix analysis to diagnose class-wise errors and failure modes.

CERTIFICATIONS

TensorFlow Developer Certificate

02/2024

Google

Seoul, South Korea

Deep Learning Specialization

10/2023

Coursera

Seoul, South Korea

SKILLS

- **Programming Languages:** Python, R, SQL
- **Optimization:** Gurobi, OR-Tools, Linear Programming, Mathematical Modeling
- **Data Science & ML:** PyTorch, Scikit-learn, TensorFlow
- **Data Engineering:** Apache Airflow, Apache Spark
- **Databases & BI:** MySQL, PostgreSQL, Tableau
- **Tools:** AWS, Git, LaTeX, Markdown

HONORS AND AWARDS

Graduated with Distinction

05/2022

University of Wisconsin–Madison

Madison, WI

Dean's List (4 semesters in a row)

2019 – 2022

University of Wisconsin–Madison

Madison, WI

Last updated: June 30, 2026